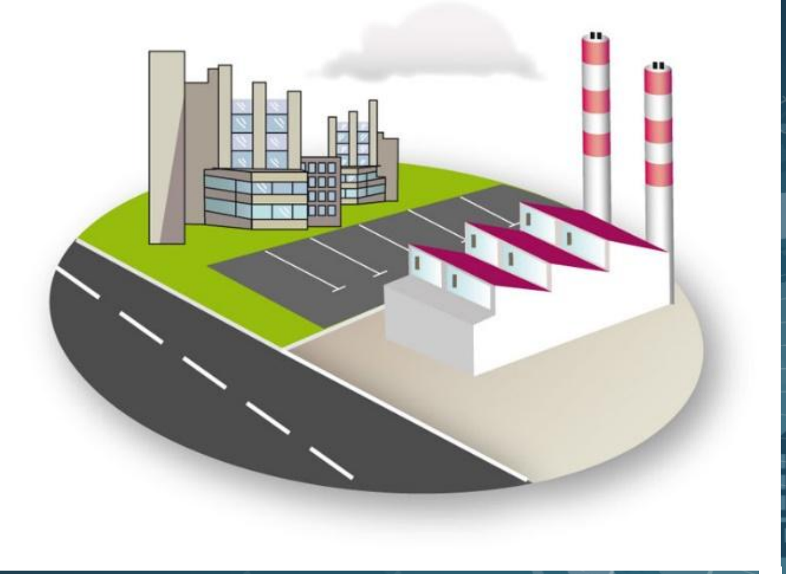

- INDUSTRIAL ORGANIZATION -

Value Stream Mapping

Course goals

- Understand what is the Supply Chain and what are the main issues of Supply Chain management.
    - Understand the objectives and benefits of Value Stream Mapping.
        - Be able to perform a VSM and to build a current state map for a product or a family of product.

Summary

1. SUPPLY CHAIN MANAGEMENT GOALS
2. OPTIMIZE SUPPLY CHAIN WITH VSM
3. VSM METHODOLOGY AND EXAMPLE

1. SUPPLY CHAIN MANAGEMENT GOALS

Definitions from APICS dictionary

- **Supply Chain :** The global network used to deliver products and services from raw material to end customers through an engineered flow of information, physical distribution, and cash
- **Supply Chain Management :** The design, planning, execution, control and monitoring of supply chain activities with the objective of creating net value, building a competitive infrastructure, leveraging worldwide logistics, synchronizing supply with demand, and measuring performance globally.

source :AIMS UK

Historical perspective

In the past, many companies placed most of their attention on the issues that were internal to their companies.

They already had suppliers, customers, and distributors, but those entities were often viewed as outside entities. Supplier were even considered as adversaries who only wanted to maximize their profit.

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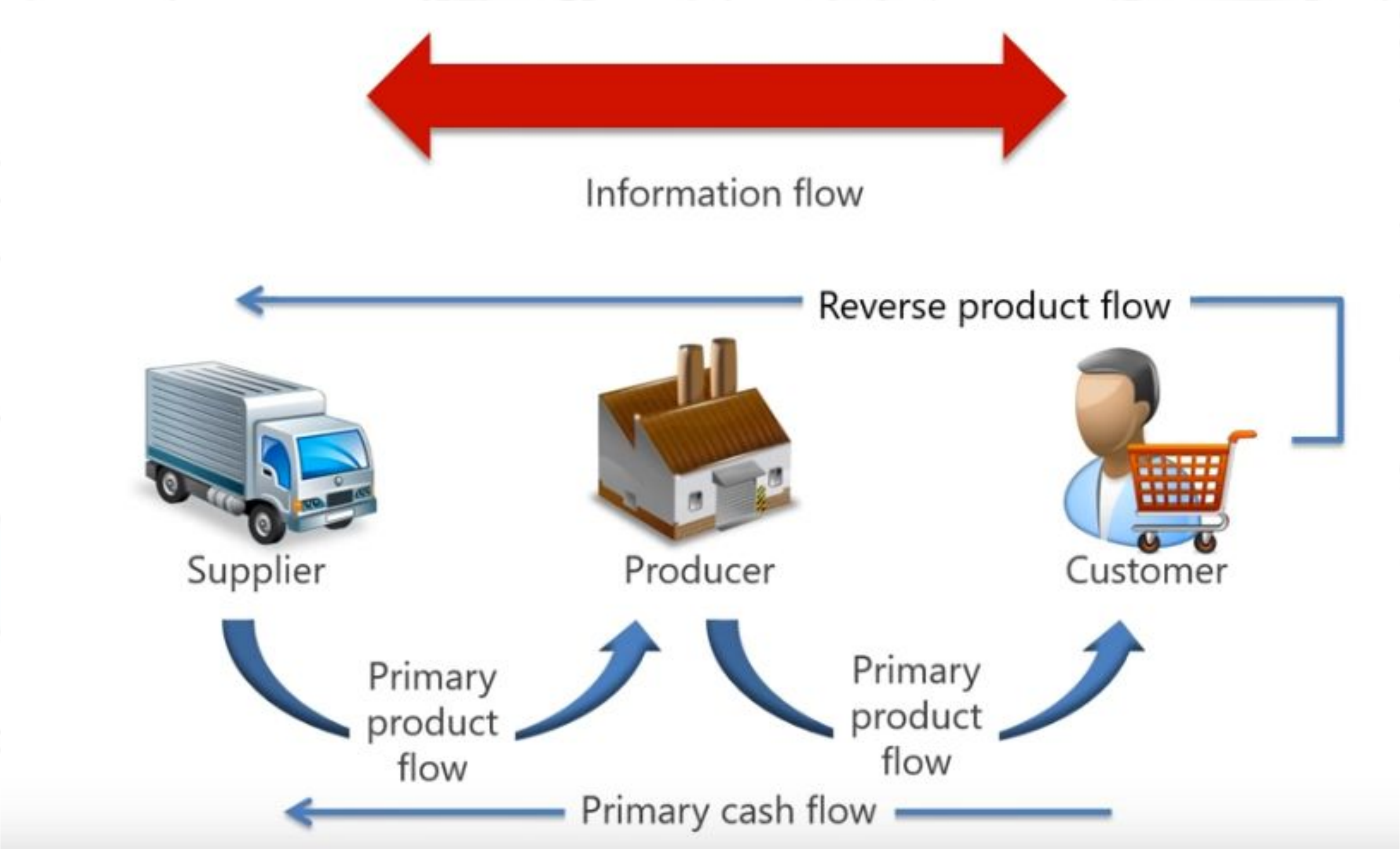
We started to see everything as a whole.

Supply Chain flows



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Supply Chain costs

Managing Supply Chain means managing the flows while optimizing the costs :

- Facility costs
- Inventory costs
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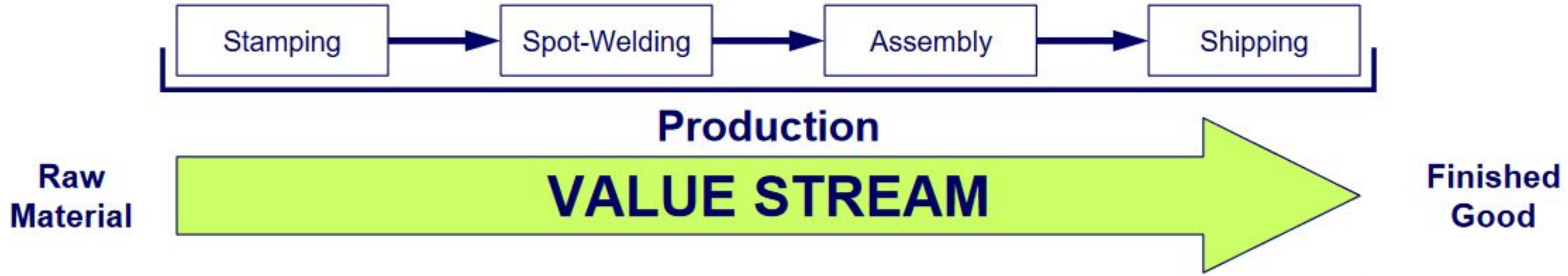
The main Supply Chain management objectives are :

- **creating value** to the customer : it is what the customer will pay for,
- **creating profit** for the company if order to make the business grow,
- Deliver the customer **at the right moment and with the good products**.

2. OPTIMIZE SUPPLY CHAIN WITH VSM

Value stream

It is all activities, both value-added and non value-added, required to bring a product from raw material to a end product in the hands of the customer.



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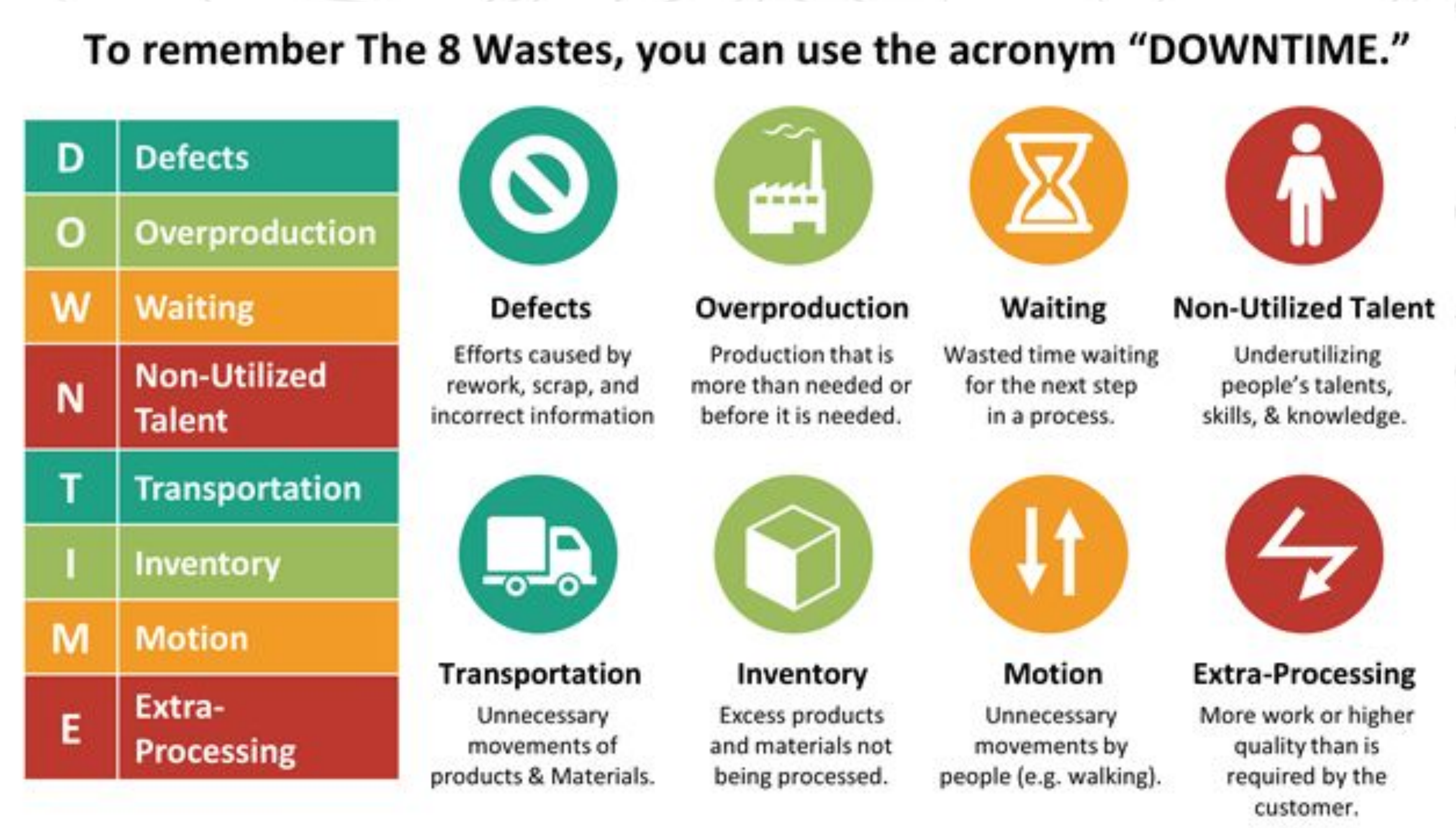
VSM should be the base of any Lean transformation in a company, as it permits to establish where it should be focused and directed.

Identify not adding-value steps



Benefits of VSM

Value stream mapping is critical for business sustainability:

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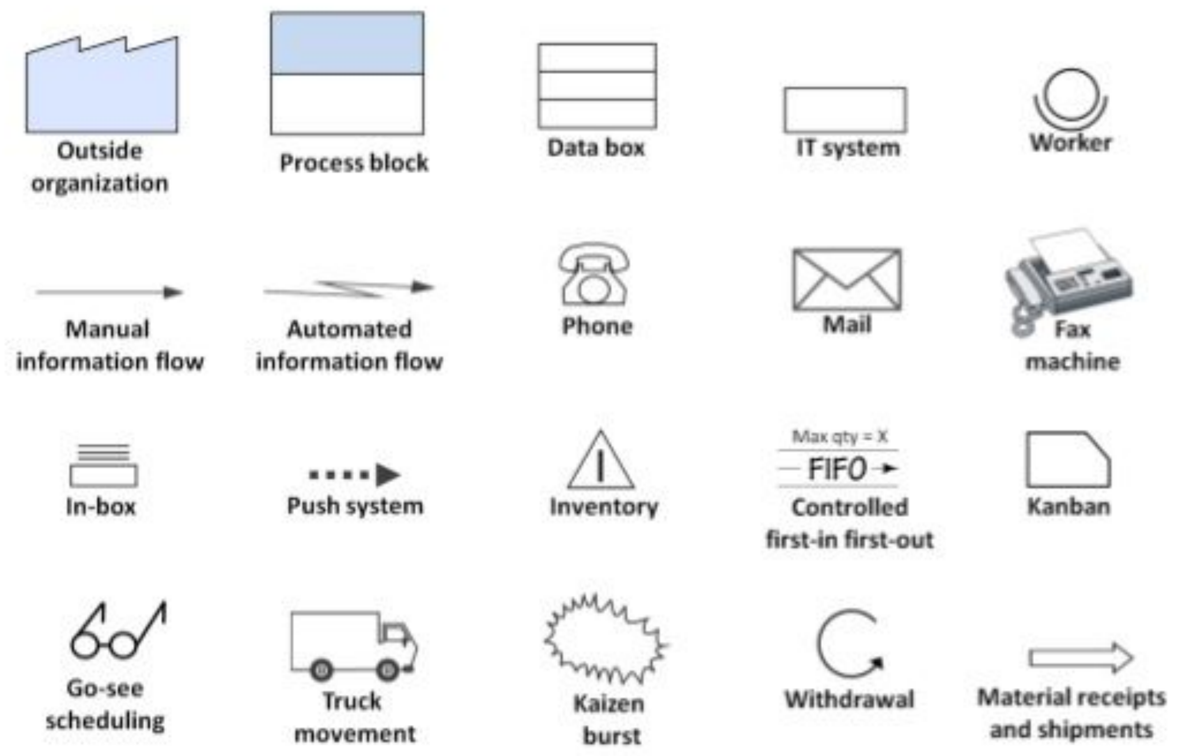
- Once wasteful handoffs are identified, your teams can consciously improve behavior, culture, communication, and collaboration.
- **Teams discard individual opinions and prioritize based on the customer's perspective.**

3. VSM METHODOLOGY AND EXAMPLE

VSM: how to build the current map

1. **Select the product** or product family to map
2. Define the **process boundaries**
3. Identify and plot the **process steps**
4. Add the **material flow**
5. Add the **inventory information**
6. Collect the **process data**
7. Add the **information flows**
8. Create the **timeline**
9. **Interpret** the data
10. Create the **future state map**

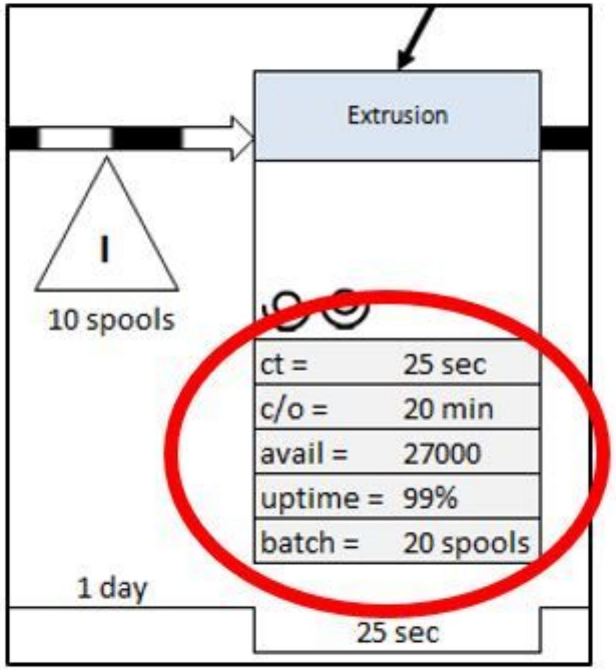
VSM icons


Outside organization
Process block
Data box
IT system
Worker
Manual information flow
Automated information flow
Phone
Mail
Fax machine
In-box
Push system
Inventory
Max qty = X
FIFO
Controlled first-in first-out
Kanban
Go-see scheduling
Truck movement
Kaizen burst
Withdrawal
Material receipts and shipments

Process data in VSM

- Number of operators
- Cycle time
- Changeover time
- Operators waiting time
- Work in process : stock before and after the process
- Machine reliability (or failure rate)
- Quality rate
- Number of accidents
- ...

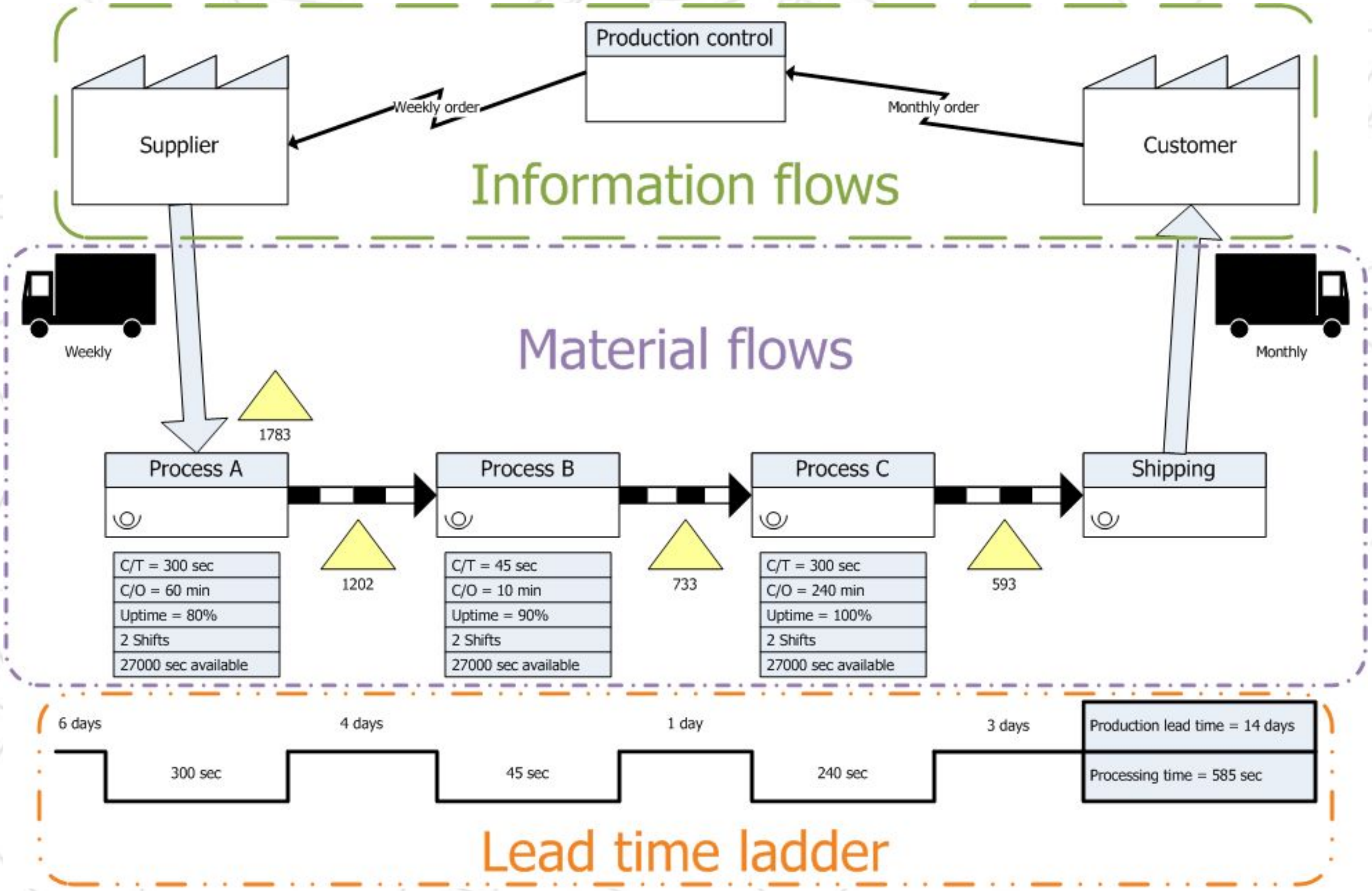
VSM DATA BOX TEMPLATE



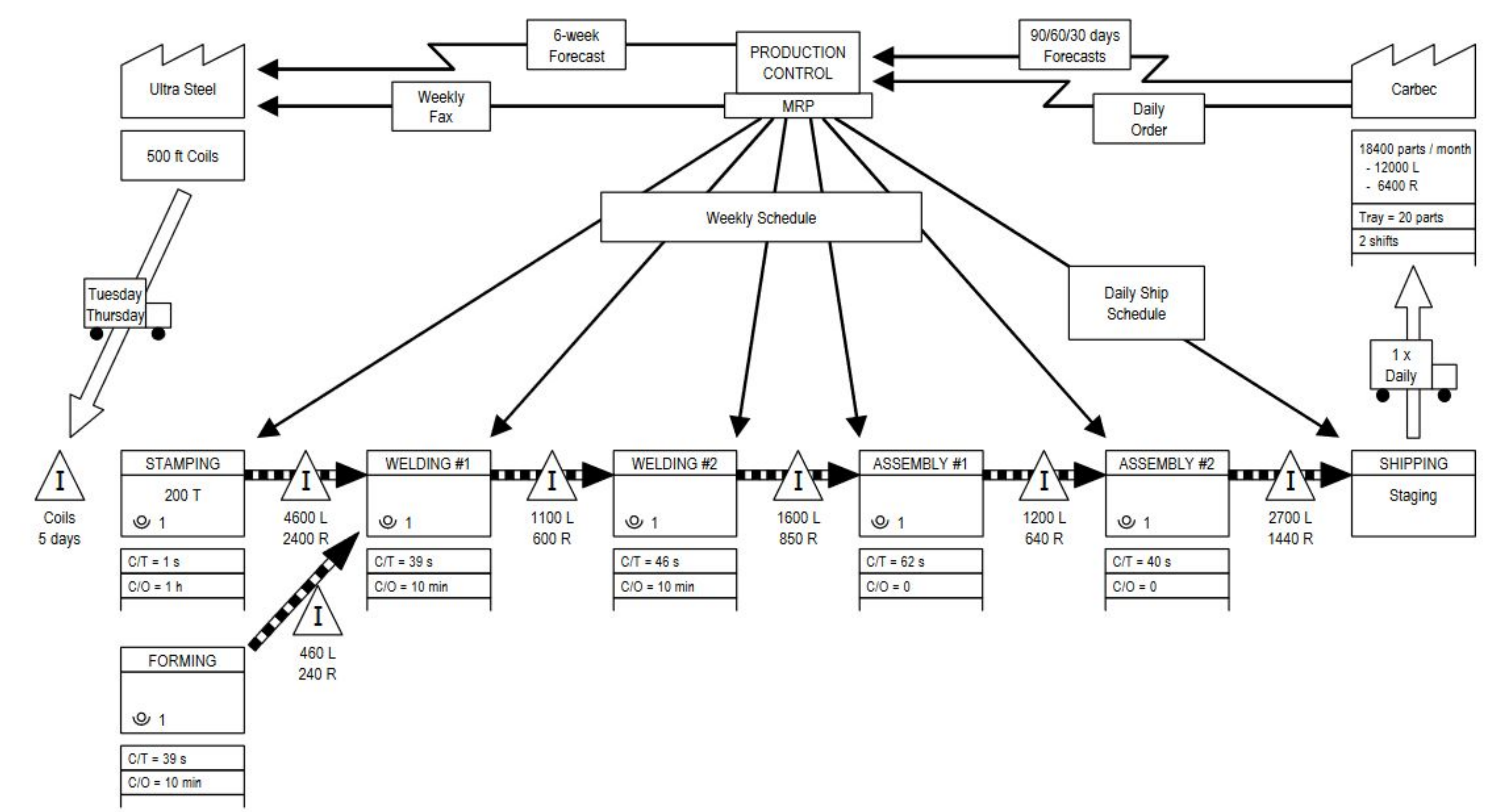
Information flow in VSM

- Forecasts from the customer (what he will buy to us)
- Forecasts to the supplier (what we will buy to him)
- Orders from the customer
- Order to the supplier
- MRP schedule
- Shipping schedule
- ...

VSM example : current state map



VSM example : current state map



VSM example : current state map

ABC Stamping Company (ABC) produces several components for vehicle assembly plants. This case concerns **one product family : a steel instrument in two types** - one each for left-hand and right-hand drive versions of the same vehicle model.

These components are sent to the **Carbec assembly plant, the Customer**.

Customer Requirements:

- **18400 parts per month** : **12000 of Type "L"** (Left-Hand drive) and **6400 of Type "R"** (Right-Hand drive).
- **One daily shipment** to the assembly plant by truck.

Work Time:

- **20 days in a month in 2 shifts** operation in all production departments.
- **8 hours every shift**, with overtime if necessary.
- Two 10-minute breaks during each shift.

VSM example : current state map

Production Processes:

- ABC's process for a product family involves **stamping** a metal part followed by **welding** and **assembly**. The components are then **shipped** to the Carbec vehicle assembly plant on a daily basis.
- Switching between Type "L" (Left-Hand drive) and Type "R" (Right-Hand drive) brackets requires **1 hour changeover in stamping** and **10-minute fixture change in the welding** processes.
- Steel coils are **supplied by Ultra Steel Company**. **Deliveries** are made to ABC Company on **Tuesdays and Thursdays**.

VSM example : current state map

ABC Production Control Department:

- Receives **Carbec's 60-days forecasts** and enters them to MRP.
- Issues ABC **6-week forecast to Ultra Steel Company** via MRP.
- Secures coil steel by **weekly Faxed order release** to Ultra Steel Company.
- Receives **daily firm order** from Carbec.
- Generates **MRP-based weekly departmental requirements** based upon Customer order, WIP inventory levels, Finished Goods inventory levels, and anticipated scrap and downtime.
- Issues **weekly production schedules** to Stamping, Welding, and Assembly processes.
- Issues **daily shipping schedule** to Shipping Department.

VSM example : current state map

Process information :

STAMPING

- Process with one operator.
- Automated 200 ton press with coil (Automatic material feed).
- **Cycle Time : 1 second** (60 parts / minute).
- **Changeover Time : 1 hour**.
- **Machine Reliability : 85%**.
- Observed WIP :
  - **5 days of coils before stamping.**
  - **4600 stamped parts of Type "L" and 2400 stamped parts of Type "R".**

VSM example : current state map

Process information :

WELDING

- Manual Process with 1 operator.
- **Cycle Time : 39 seconds.**
- **Changeover Time : 10 minutes.**
- **Reliability : 80%**
- Observed WIP : **1100 parts on Type "L" and 600 parts on Type "R".**

ASSEMBLY

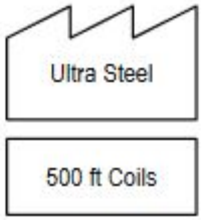
- Manual Process with 1 operator
- **Cycle Time : 62 seconds.**
- **Changeover Time : None.**
- **Reliability : 100%**
- Observed WIP : **1200 parts on Type "L" and 640 parts on Type "R".**

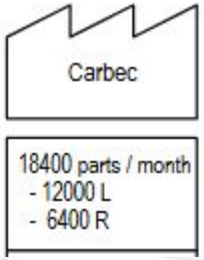
SHIPPING DEPARTMENT: Removes parts from finished goods warehouse and stages them for truck shipment to Customer.

VSM example : current state map

1st step : Process boundaries

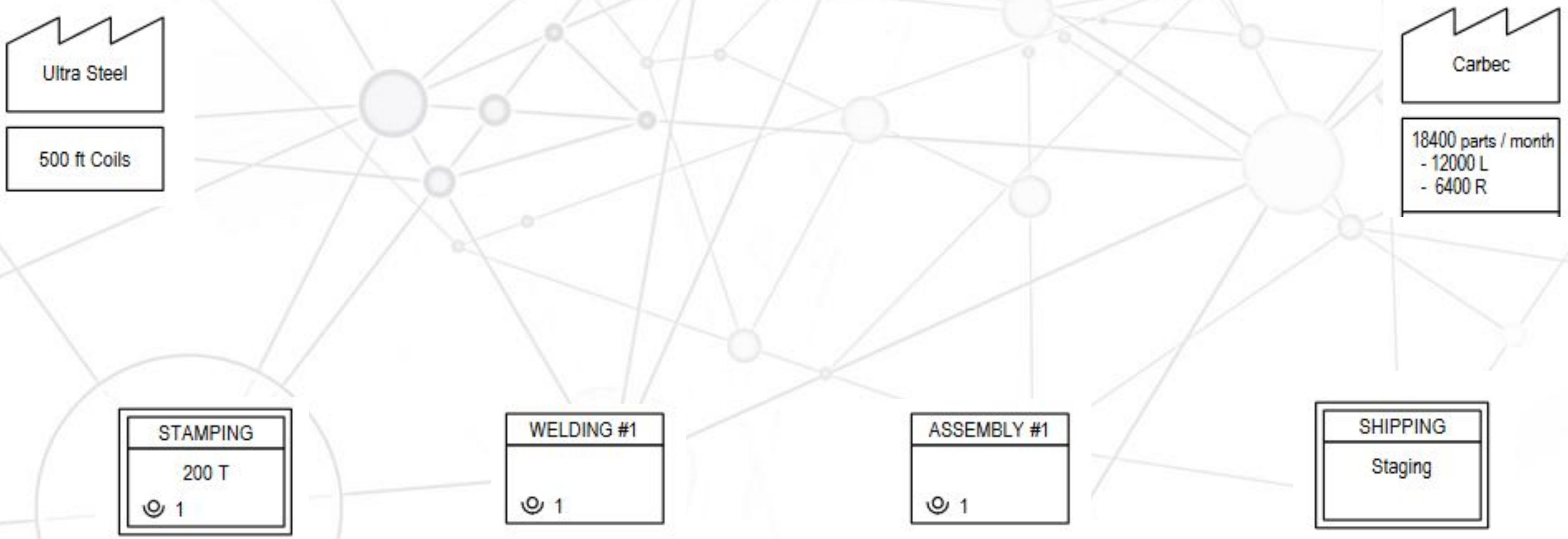
Where does the value stream starts and ends in this particular case?





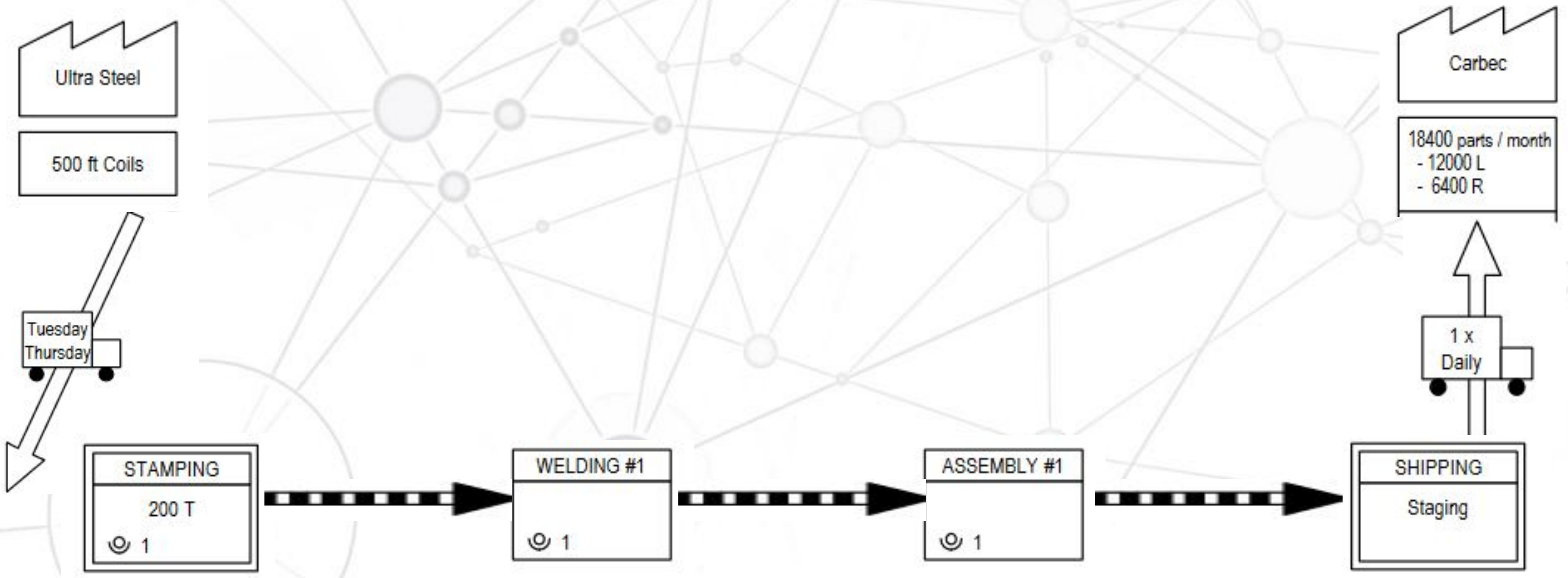
VSM example : current state map

2nd step : Identity and plot the process steps



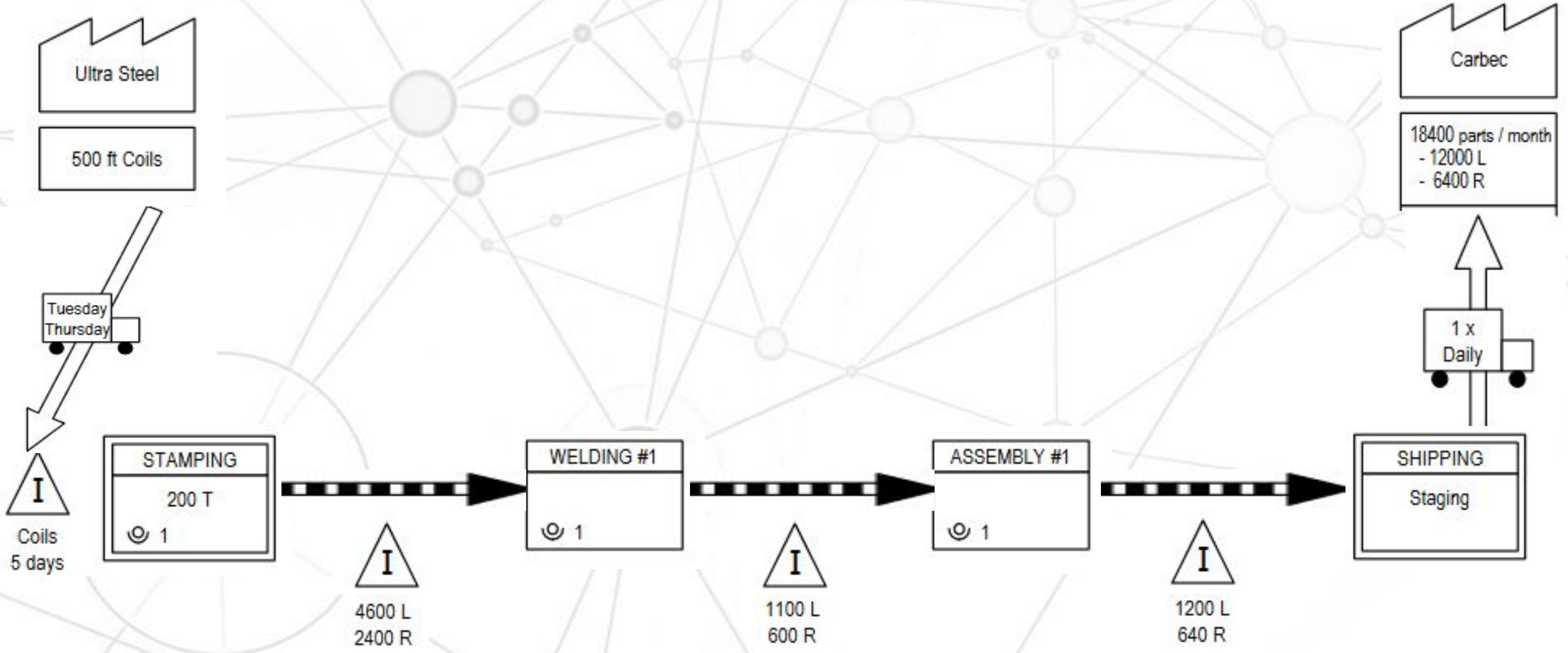
VSM example : current state map

3rd step : Add the material flow



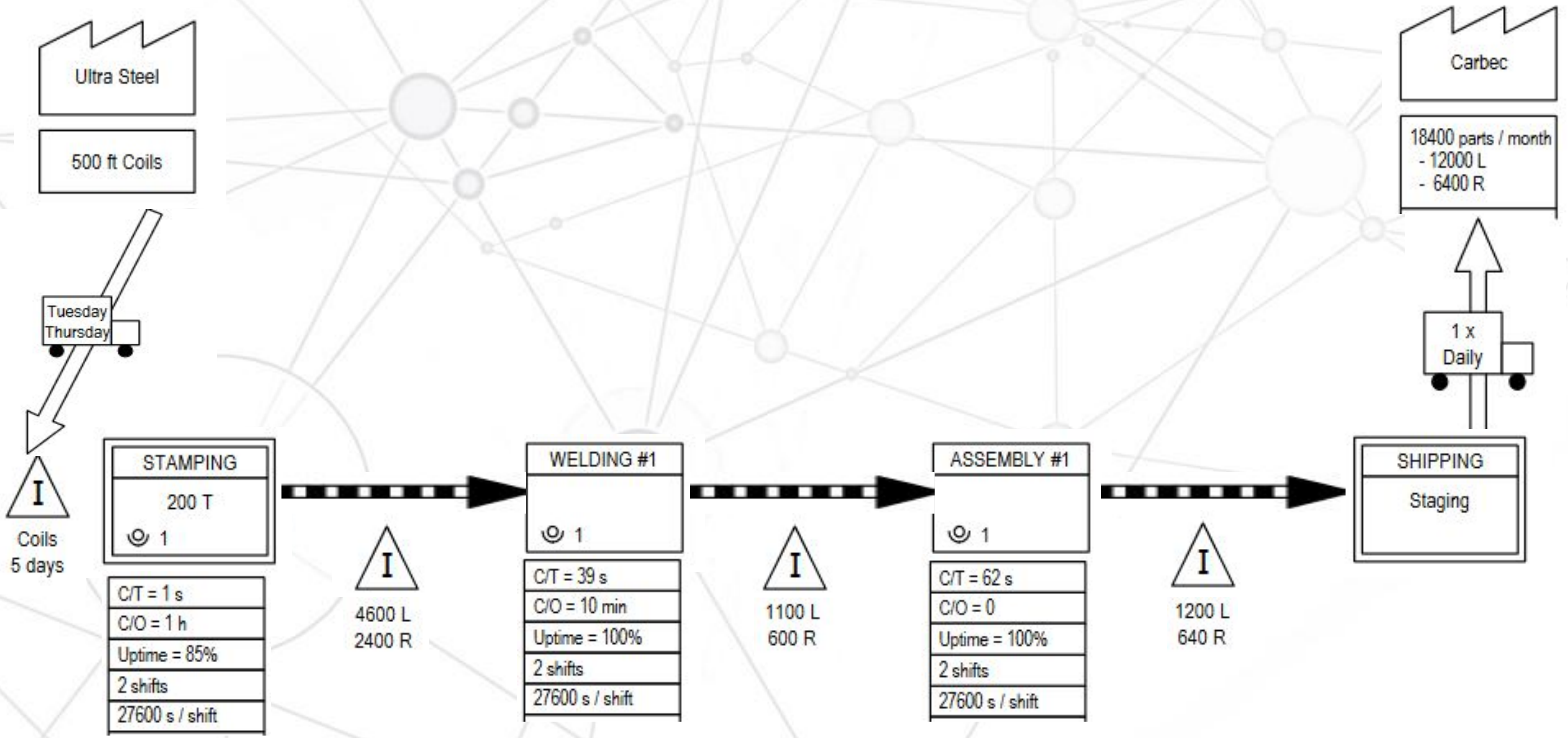
VSM example : current state map

4th step : Add the inventory information



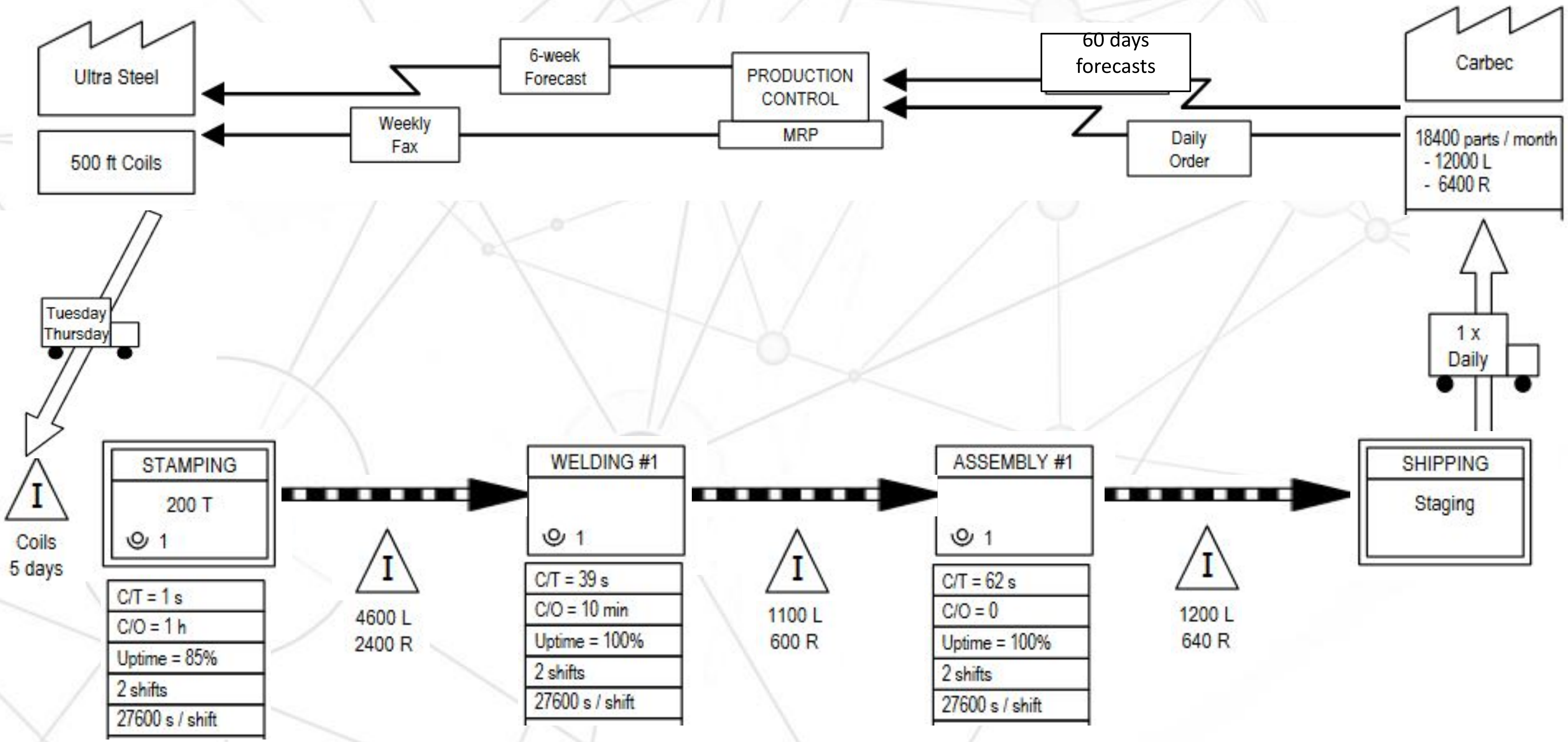
VSM example : current state map

5th step : Add the process information



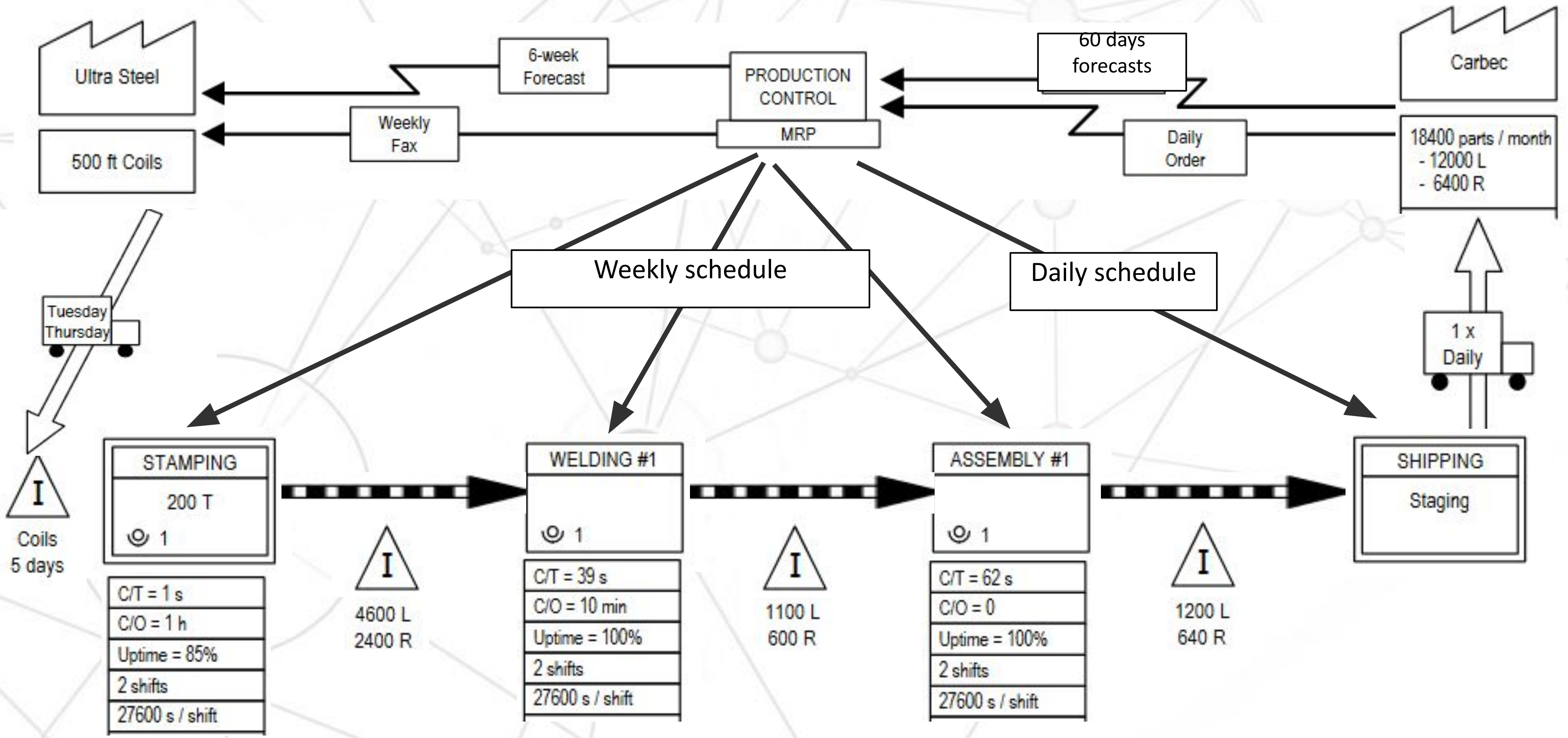
VSM example : current state map

6th step : Add the information flow



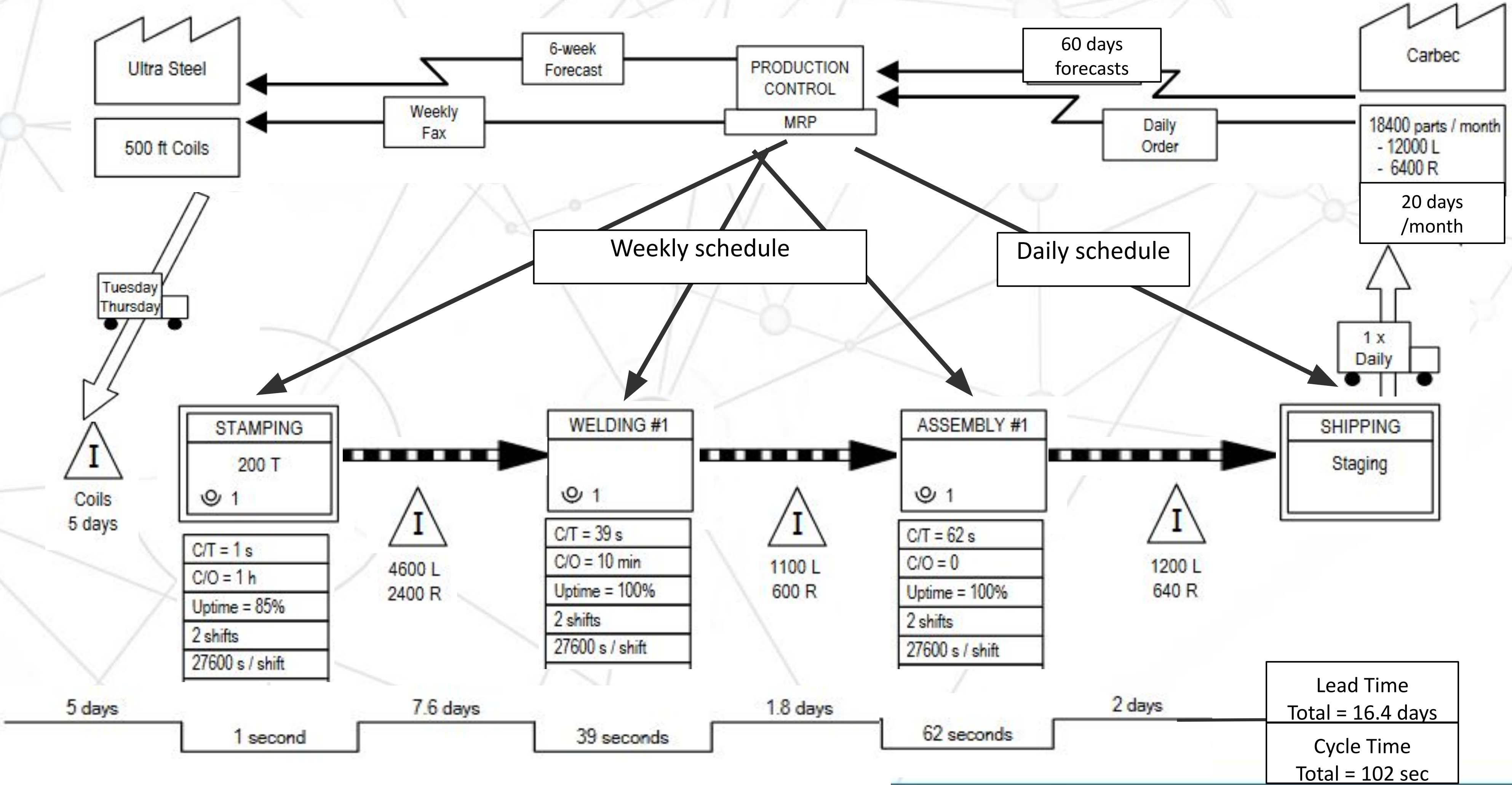
VSM example : current state map

7th step : Add the information flow



VSM example : current state map

8th step : Add the timeline



The future state map

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Basically, we should try when it is possible to :

- Align the production rate on the pace of the market (Takt Time)
- Use pull flows instead of push
- Build continuous flow and build stock only if necessary

Thanks for your attention

DO YOU HAVE QUESTIONS?

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